MediaPipe with LSTM Architecture for Real-Time Hand Gesture Recognization*

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Abstract. Gesture recognition plays a vital role in the area of research for human-computer interaction (HCI). The integration of MediaPipe with Long Short Term Memory (LSTM) architecture holds tremendous potential for real-time hand gesture recognition. MediaPipe provides a robust and versatile framework for capturing and processing multimedia input, such as video streams from cameras or pre-recorded video files. The temporal modeling capabilities of LSTM captures the temporal dynamics of hand gestures. This research paper aims to present a novel method utilizing the MediaPipe with LSTM architecture for real-time hand gesture recognition. A test on real-time gesture recognition is performed to evaluate the performance of the suggested model. Our results demonstrate that the suggested method outperforms other state-of-the-art approaches on our custom made dataset with an accuracy of 98.99%.

Keywords: Gesture recognition · LSTM · MediaPipe · Human-computer interaction

1 Introduction

Gesture recognition is a way to read and understand a person's body language to communicate effectively. Based on computer vision technology, hand gesture recognition is getting great attention because of it's natural way of interaction between humans and computers. Different hand poses and their movements generally make a gesture [1]. Real time hand motion detection also depends on the image processing capability of the system and hand movements. In some cases, if the hand movements are too fast then it is a difficult task to capture the appropriate hand motions. So, figuring out the position of a hand in real time is

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still a difficult task that requires work from many different fields, such as pattern recognition, computer vision and image processing. A gesture is a sign for expressing one's behaviour or feelings physically. Humans and computers may communicate via the use of gestures. The intent of the user can be determined by analyzing the user's gestures and movements [2]. Multiple input images are collected to detect dynamic motion, but just a single image is needed to detect static gestures. The quality of the supplied image is improved using image enhancement techniques. Dynamic gesture recognition falls under the category of video classification since the data is predominantly presented in the form of video frames. Dynamic gesture detection is difficult because of factors such as inconsistent pixel quality, camera movement, and individual diversity in the execution of a particular action. It is difficult for an algorithm to accurately predict gestures due to the context dependent nature of gestures and the continuous movement of hands and arms. Overall, the integration of MediaPipe with LSTM architecture offers a powerful solution for real-time hand gesture recognition [3]. The subsequent sections of this paper are structured as follows: In Section 2, we discuss the deep learning techniques that other researchers have implemented for gesture recognition. The approach behind the MediaPipe based LSTM architecture for recognizing hand gestures in real-time is outlined in Section 3. Experiment results are described in Section 4. The last component of the paper serves as the final chapter.

2 Related Work

Hand gesture recognition is an essential aspect of HCI with significant implications for a wide range of applications, including virtual reality, robotics, sign language interpretation, and smart homes. Several methods are developed in this area including template matching, decision trees, and deep learning, to recognize hand gestures. Numerous approaches and techniques are proposed for hand gesture recognition. Throughout the years the system's accuracy and performance are improved in recognizing hand movements. In recent years, numerous studies are conducted utilizing MediaPipe and LSTM models for hand gesture recognition. Several research projects have used MediaPipe to analyze hand gestures. Javier et al. [4] proposed a framework for motion based head and eye gesture detection using egocentric visual input from an eyewear device. The approach is predicated on a LSTM for storing visual associations across time, and a homography estimation convolutional neural network (CNN) for representing motion between frames. In order to capture the temporal visual correlations the CNN output is fed into an LSTM network. The system operates with an accuracy of 90%. between 20 and 30 frames per second MediaPipe posture is proposed in conjunction with convolutional neural network (CNN) models for LSTM based traffic gesture detection by Ong et al [5]. In this method video data is analyzed using an LSTM classification model and a pose extraction tool MediaPipe. This allows for the detection and categorization of traffic movements. The Mediapipe posture recognition approach provides very accurate predictions of 3D body key-

point positions in real time. All kinds of gestures can be reliably classified by the model. A deep learning-based approach was introduced by Areeb et al. [6] to achieve precise prediction of Indian Sign Language (ISL). After the annotation of frames two models are created: one for object detection and another for categorization. One model combines traditional RNNs with a long short-term memory (RNN-LSTM) strategy, while another makes use of 3D convolutional neural networks (CNNs), VGG-16, and a LSTM network. The latest model was developed on top of the advanced YOLOv5 (You Only Look Once) object identification algorithm. The prediction accuracies of the two classification models are 82% and 98%, respectively with an average accuracy of 99.6%. The test results shows that the YOLO based model outperformed its competitors. Sundar et al. [7] suggested a technique for recognising ASL alphabets that is able to distinguish between static and dynamic motions in American Sign Language (ASL). Google's Mediapipe is used to acquire hand landmarks and a custom dataset is built for testing. Successful hand gesture recognition is achieved with the help of LSTM. The proposed approach is evaluated using 26 alphabets, and it achieves 98.99% accuracy. Recent research using MediaPipe and LSTM models for hand gesture recognition has shown promising results. Accurate and fast methods for hand identification and landmark estimation are provided by these approaches and also captures sequential relationship between hand motions.

3 Methodology

In this research, we provide a deep learning model for real-time gesture recognition using LSTM and the Mediapipe framework. Hand detection, landmark estimation, and gesture recognition [8] are the three basic pillars of the suggested technique. MediaPipe is used for hand detection and landmark estimation, while LSTM models are used for hand gesture recognition. The system can reliably recognize and follow the hand in real time due to MediaPipe's pretrained algorithms for hand identification and landmark estimation. The system uses a pretrained MediaPipe model to automatically extract the hand's landmark points [9] for successful hand detection and tracking. The landmark points are then used to train LSTM models to recognize gestures based on patterns of landmark points. The technique makes use of motion based signals, therefore the data received from video streams of signs should match the defined activities. In order to extract information from a moving sign, it is necessary to gather and combine data from each frame of the video in sequential sequence. Movement based signal data is significantly important to identify hands and body posture [10]. The human skeleton's joints serves as reference for these hand and body components. Each sign will exhibit a unique configuration of these core characteristics based on the behaviours it represents. An LSTM based deep learning strategy [11] is used to predict the sign from this pattern. The suggested hand gesture recognition system's flowchart is shown in Fig. 1.

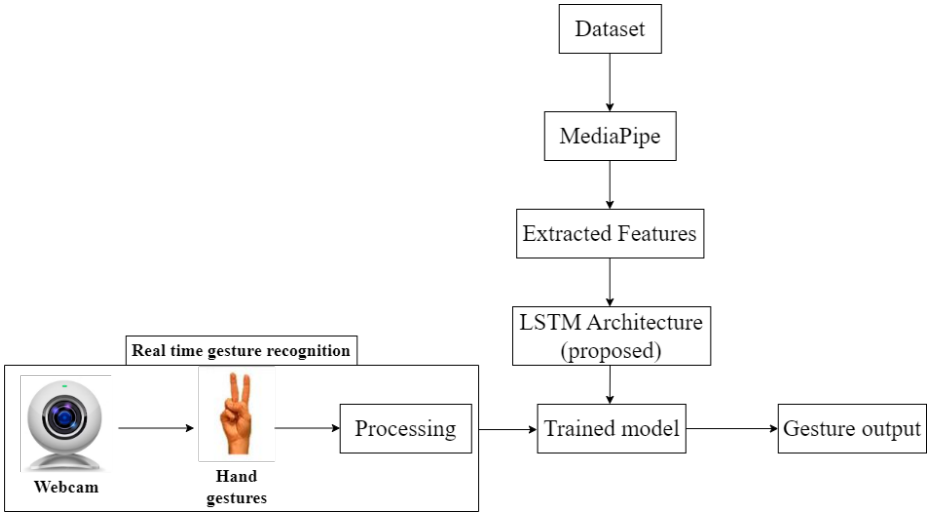

Fig. 1. Training and testing flow diagram of the proposed architecture

3.1 Dataset Description

In this research, a database of hand and facial gestures for five distinct activities ("Hello", "Help", "Thanks", "Go to" and "Bathroom") is compiled. For each gesture a total of 200 images are taken with face, hand and body motion. The lighting, and visibility factors while recording the face, hands, and body are maintained as natural environmental conditions. The performance of the Mediapipe model can degrade significantly for non-frontal faces, low-resolution images, and for hands in low-resolution scenarios. So, an average height of 5.7 feet human is considered for this experiment. The final data set is then separated into a 90% training set and a 10% testing set. All of the testing is done on a machine running Windows 10 and equipped with a 10th generation Intel processor and an NVIDIA GTX 1650TI graphics card with 4 GB of RAM. For the purpose of training networks, the popular PyTorch framework has been deployed. Python 3.10.9 and the Pytorch 1.13.1 framework were used in the experiment.

3.2 Data Preprocessing

The dataset is divided into its constituent activities during the data preparation phase, with 200 frames of images from each video that represents each individual activity. The MediaPipe library is used to identify the subject's keypoints like pose, face, left hand, and right hand in each image. From each frame, the most important characteristics with these four important keypoints are extracted. For each action, we compiled 200 data point sets with the accompanying keyframes. We used a total of 1000 data sets for model development and evaluation. This procedure of gathering data and determining its significance is repeated for each

activity in the dataset. After that, landmark information is extracted from the images through MediaPipe to identify key anatomical features of the hand, body, and face. The MediaPipe holistic model [12] returns landmarks for hand, body, and face from each single frame. These are known as x, y, and z coordinates. Visibility is a further consideration in body posture. There are 21 individual controls for each hand. Identifying a posture requires a total of 33 identifiers. The human face has 468 identifiers that may be used to decode emotions. This implies that the number of coordinates obtained was 1662 for a single frame (21*3 for each hand on the left and right, 33*4 for the body position, and 468*3 for the face). A dataset of shapes (200,1662) was created by stacking a total of 1662 landmarks, including the subject's face, attitude, left hand, and right hand. The number 200 denotes the number of frames in an action, and the number 1662 denotes each landmark dataset size. The coordinates and landmark keypoints

Table 1. The defined keypoints and coordinates

Body Parts	Coordinates	Keypoints
Right Hand	63	21
Left Hand	63	21
Face	1404	468
Body Pose	132	33
Total	1662	543

extracted through MediaPipe for different body parts are given in Table 1.

3.3 The proposed LSTM network Architecture

Training costs and error margin minimization in the output is the primary focus of LSTM model. The LSTM models [13] use the time-dependent relationships in hand movements by using the sequential information of the landmark points. The model is trained for 120 epochs on the training data, using a batch size of 13 for faster convergence during training and to lead better generalization. The LSTM models are trained using a collection of hand gesture videos that include both static images and dynamic images. LSTM models [14] are trained on the training set using the Adam optimizer with a learning rate of 0.001. The input video sequence to the LSTM is the MediaPipe extracted keypoints [15] with the shape of (200, 1662). The novel neural network architecture is designed with four layers of networks. The first two layers are LSTM layers and last two layers are dense layers. The model structure is shown in Fig. 2. To capture complex patterns in sequential data the Layer 1 of a Long Short-Term Memory (LSTM) network [16] is modified with 64 neurons and uses sigmoid activation with the parameter 'return_sequences' is set as 'True' for sending input to the Layer 2 of the LSTM layer from layer 1. If the "return_sequences" is not set as 'True' then the layer 2 will not receive input from layer 1. The LSTM layer 2 has 128 neurons and sigmoid as an activation function. This modified deep LSTM

architecture can capture hierarchical features and dependencies in the data. The hierarchical network structure is created with the combination of two dense layers. This allows for a combination of temporal sequence modeling (LSTM) and feature extraction (dense). The first dense layer uses 64 neurons with the ReLU activation function. The last dense layer is the output layer uses softmax activation function and has number of neurons same as number of gesture [17] classes. The number of cells in the output layer is set as 5 the represents the number of classes. On the testing set, the models are evaluated by the evaluation measure [18]. The suggested model's success is measured by its accuracy [19].

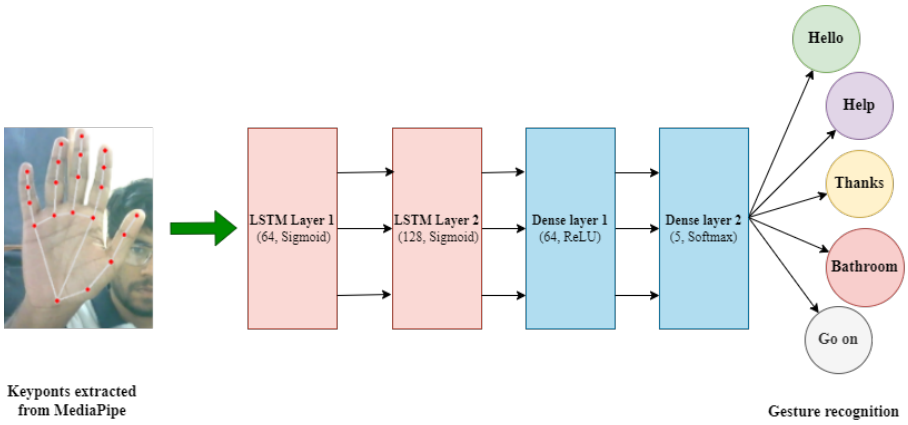

Fig. 2. The suggested LSTM model's network with pair of LSTM layer and Dense layer.

4 Result Analysis and Discussion

Models are trained and processed using our custom build dataset to perform precise gesture recognition. Fig. 3 shows a variety of gestures that are recognized with our proposed model including "Hello", "Help", "Thank you", "Bathroom" and "GoTo". In the images, coloured dots stand in for landmarks and these points of interest are associated with coordinates (x, y, and z) and visibility levels. The most important x, y, and z positions are face, left hand, and right hand and with this the pose visibility value is also considered. In this research, we make use of the MediaPipe library to provide a framework for gesture recognition. The test results shows that the quality of MediaPipe's gesture recognition module creates a significant impact on the precision of recommended models. We also compared the suggested LSTM architecture of hand gestures to the state-of-the-art LSTM model after training the dataset on the proposed design. Model 1 describes the performance accuracy of the proposed LSTM architecture, whereas